10 The table gives some information about the times taken by 80 students to complete a test.

Time (t minutes)	Frequency
$20 < t \leqslant 25$	4
$25 < t \leqslant 30$	16
$30 < t \leqslant 35$	30
$35 < t \leqslant 40$	18
$40 < t \leqslant 45$	12

(a) Complete the cumulative frequency table for this information.

Time (t minutes)	Cumulative frequency
$20 < t \leqslant 25$	4
$20 < t \leqslant 30$	20
$20 < t \leqslant 35$	50
$20 < t \leqslant 40$	68
$20 < t \leqslant 45$	80

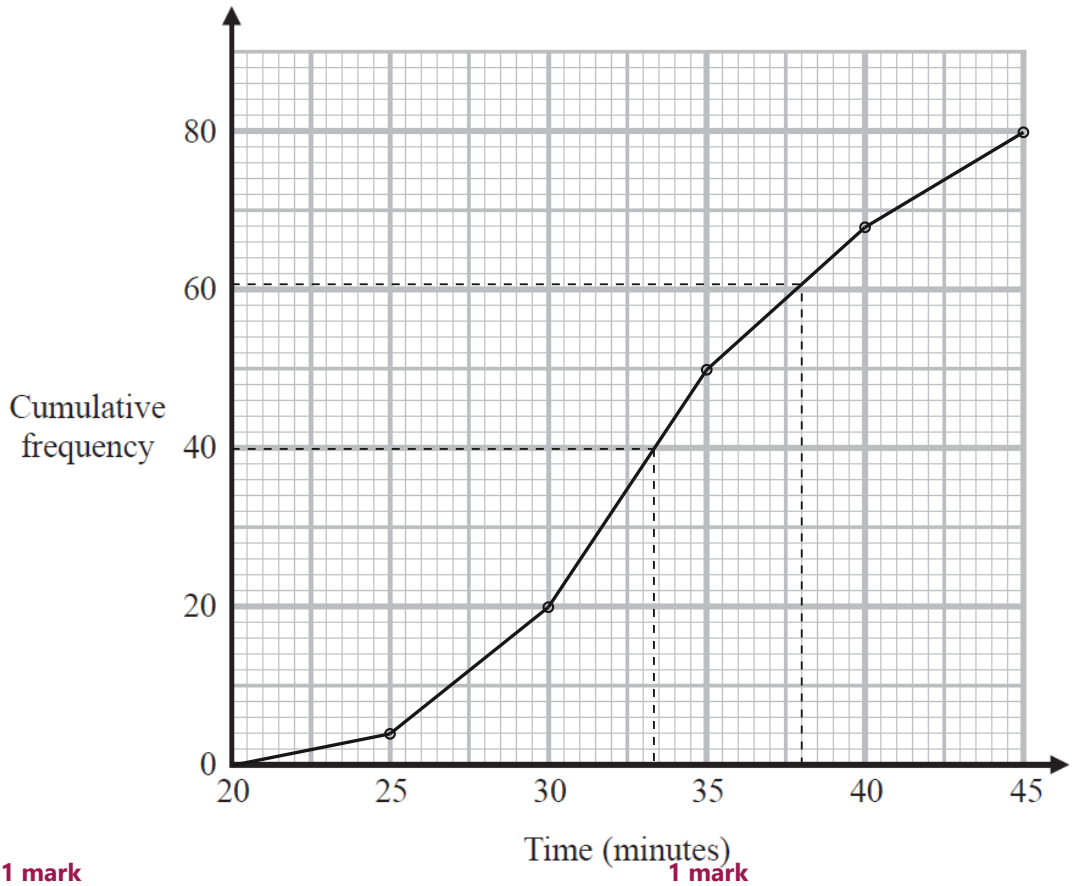
1 mark
Running totals: 4, 20, 50, 68, 80.

(1)

(b) On the grid opposite, draw a cumulative frequency graph for your table.

Plot at each upper class boundary:

(25,4), (30,20), (35,50), (40,68), (45,80)



1 mark
Plot upper boundaries.

1 mark
Join increasing points.

(2)

(c) Use your graph to find an estimate for the median.

Median: 40th student → about 33 minutes.

1 mark
Read from CF = 40.

..... minutes
(1)

One of the students is chosen at random.

(d) Use your graph to find an estimate for the probability that this student took longer than 38 minutes to complete the test.

At 38 minutes, cumulative frequency ≈ 61.

1 mark
Subtract from 80.

Longer than 38 minutes: 80 − 61 = 19.

Probability $\approx \frac{19}{80} \approx 0.24$

1 mark
About 0.24.

.....
(2)

(Total for Question 10 is 6 marks)